

Improving Colonoscopy Reconstruction

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Spring 2023

1 Introduction

Colonoscopy is a widely used diagnostic procedure in the field of gastroenterology that allows physicians to visually examine the colon for abnormalities, lesions, and signs of disease. It plays a crucial role in the early detection and treatment of colorectal cancer, inflammatory bowel disease, and other gastrointestinal conditions. However, the accuracy of diagnosis and subsequent treatment heavily relies on the quality of colonoscopy reconstruction.

Colonoscopy reconstruction involves the creation of a three-dimensional (3D) model or representation of the colon based on the captured video footage during the procedure. By utilizing advanced algorithms and methodologies, we aim to overcome challenges related to image quality, occlusions, and tracking issues encountered during colonoscopy procedures. In this report, we will explore two computer vision-based projects and our conclusion: CNN SLAM and BlueLight Reconstruction.

2 CNN SLAM

CNN SLAM is a technique that combines Convolutional Neural Networks (CNNs) with Simultaneous Localization and Mapping (SLAM) to reconstruct the colon from colonoscopy videos. The primary objective of CNN SLAM in colonoscopy reconstruction is to create accurate 3D reconstructions of the colon by leveraging the visual information captured in the video frames. The algorithm processes each frame, extracts relevant features, and simultaneously estimates the camera pose and generates a dense point cloud representation of the colon surface.

This framework was already developed and utilized by the group, however I was responsible for running the CNN SLAM algorithm over entire colonoscopies. This entailed running an entire video until it failed and focusing on successful video segments known as "chunks" before discarding this "chunk" and repeating this processes on the remaining video. Most "chunks" typically range from 10 frames to about 200 frames. For this experiment I focused on the following colonoscopy: **Auto_A_Sep15_12-48-02.mov**

The CNN SLAM takes an input of GeoDepth which can be seen above. As we can see in **Figure 1**, the GeoDepth and put in sometimes more accurate than others. On the far left, the corresponding depth input seems fairly reasonable; however on the far right, the depth map is far less convincing.

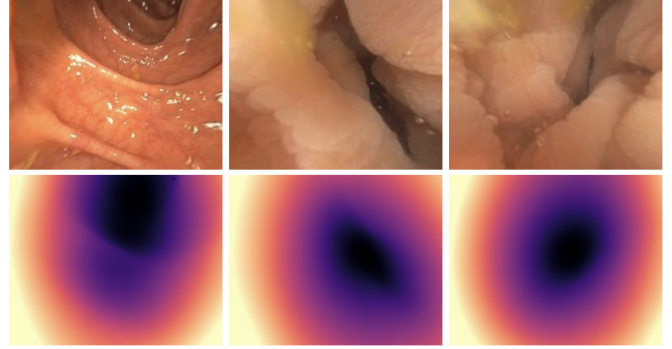


Figure 1: Depth Input The top row are randomly selected frame from the colonoscopy and the bottom row are three corresponding depth maps. While and yellow represent parts of the surface that are near the camera while black and purple represent parts of the surface that are far the camera.

During the analysis of CNN SLAM, we compared the performance of two variants: Geodepth and Geodepth + Normal. Geodepth refers to the baseline method that uses only geometric information, while Geodepth + Normal incorporates additional normal information to enhance the reconstruction.

2.1 GeoDepth

The run length of a chunk, when considering the Geodepth-only input, averaged around 54 frames. By excluding outlier chunks with lengths greater than 200 frames, the average run length was approximately 43 frames. These findings indicate that the algorithm was able to track and reconstruct the colon surface for a considerable duration, allowing for continuous reconstructions over multiple frames.

The point clouds generated by the Geodepth-only input exhibited a high level of detail. The points frequently formed planes and curves, which closely resembled the walls of the colon. Notably, there were clear curves in the point cloud that corresponded to the curves observed in the corresponding video chunk. This alignment between the reconstructed point cloud and the actual colon anatomy suggests the effectiveness of the Geodepth-only input in capturing the geometric features of the colon surface.

Furthermore, for longer runs, the point clouds displayed a tunnel or cylinder shape, indicating the ability of the algorithm to capture the three-dimensional struc-

ture of the colon. The scatter within the point cloud was generally small to medium, signifying a relatively coherent reconstruction.

Additionally, the Geodepth-only input encountered tracking issues. These issues included tracking loss, bad predictions, and initialization failures. These challenges affected the temporal consistency and accuracy of the reconstruction process, potentially leading to gaps or inaccuracies in the resulting point cloud.

2.2 GeoDepth + Normals

When comparing the run length of the Geodepth + Normals configuration to that of the Geodepth-only input, it was observed that the addition of normals led to longer run lengths. On average, the run length increased to approximately 67 frames, showcasing an improvement over the Geodepth-only input. By excluding outlier chunks with lengths greater than 200 frames, the average run length remained relatively high, at around 55 frames. This increase in run length indicated that the algorithm was able to track and reconstruct the colon surface for a more extended duration.

However, while the run length improved, the quality of the point cloud deteriorated. The point cloud generated with the Geodepth + Normals input did not exhibit clear shapes or structures. The points appeared scattered, with medium to high variability between them. In contrast to the Geodepth-only input, where some tunnel shapes were visible for longer chunks, the addition of normals did not result in significant improvements in point cloud quality, except for very long chunks exceeding 200 frames.

Furthermore, similar to the Geodepth-only input, the Geodepth + Normals configuration encountered tracking issues, including tracking loss, bad predictions, and initialization failures. However, with the inclusion of normals, tracking was able to be resumed more frequently than in the control case. This resumption of tracking contributed to a more continuous reconstruction process and provided better temporal coherence in the reconstructed frames.

2.3 Conclusions

A comparative analysis of the results obtained from RNN SLAM with Geodepth and Geodepth + Normals inputs provides valuable insights into the performance and limitations of these approaches. It is evident that the addition of normals in Geodepth + Normals configuration increases the run length of the reconstruction, while the Geodepth-only input demonstrates clearer visible shapes in the generated point cloud. However, both inputs encounter similar tracking discrepancies and exhibit failures at similar frames.

One of the primary reasons for tracking failures in both configurations is the presence of large frame jumps. These jumps occur during the pre-processing stage when a significant portion of the video is removed, leading to a sudden transition from a "down the barrel" to an "en face" view without a smooth transition. Such abrupt changes disrupt the tracking process and hinder the accurate reconstruction of the colon surface.

Another contributing factor to tracking issues is the inconsistency in lighting conditions. Close occlusions, which have higher intensity values than the distant view, result in lighting inconsistencies or transitions that occur rapidly. These rapid lighting changes often cause the tracking to break, further affecting the reliability of the reconstruction.

Additionally, preprocessing challenges such as the presence of bubbles, film, and other liquids in the colon pose obstacles to the algorithm's ability to generate an accurate reconstruction. These factors interfere with the clarity of the captured images and introduce noise that hampers the reconstruction process.

In conclusion, improving the pre-processing methods is crucial to enhancing the accuracy and reliability of the CNN SLAM algorithm. Addressing issues such as large frame jumps, inconsistent lighting, and the presence of bubbles and other liquids would significantly contribute to achieving more precise reconstructions of the colon. By mitigating these challenges, a more comprehensive evaluation of the reliability and effectiveness of the CNN SLAM method can be achieved, paving the way for improved colonoscopy reconstruction techniques in the future.

3 BlueLight

The BlueLight Reconstruction project focuses on leveraging optical flow and surface reflectance measurements for improved colon reconstruction. This technique aims to create a data-driven surface reflectance model that enhances the utility of light field information in colon visualization.

The BlueLight Reconstruction project utilizes optical flow techniques to estimate the correspondence of surface motion in colonoscopy videos. By analyzing changes in pixel intensities between frames, the algorithm calculates the apparent motion of objects within the image sequence. This information is then used to reconstruct the geometry of the colon. Additionally, surface reflectance measurements are taken into account during the reconstruction process. These measurements help to improve the accuracy of the reconstructed model by considering the reflectivity properties of different regions within the colon.

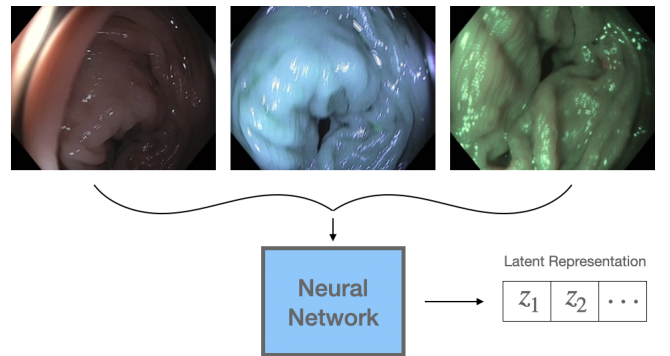


Figure 2: BlueLight Flow Chart We intend to feed a collection of frames (video) into a neural network and receive a latent space representation of the frames

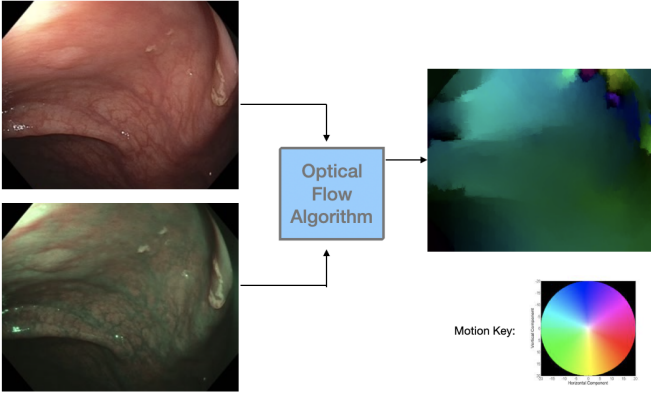


Figure 3: Optical Flow, Flow Chart we feed to sequential images into an optical flow algorithm and dance optical flow is returned. The flow indicates the direction from image 1 to image 2. This includes a key at the bottom and in this case, the features of the image tend to move to the top left.

The motivation behind BlueLight Reconstruction stems from the challenges faced in traditional colon reconstruction methodologies. These challenges include difficulties in image correspondence due to areas with low contrast or obstructions caused by fluid or other substances. By incorporating optical flow and surface reflectance measurements, BlueLight Reconstruction aims to overcome these limitations and provide more detailed and accurate reconstructions of the colon. Currently, we are just extracting the albedo which is a vast oversimplification of the surface reflectance.

We first started by implementing a non-learning algorithm to determine if optical flow was a viable approach for obtaining surface reflectance. Later, given the success of this non-learning algorithm, we are moving towards implementing a unsupervised neural net work to perform this task.

3.1 Optical Flow

Optical flow is a computer vision technique used to estimate the apparent motion of objects within an image sequence. It provides a dense vector field that represents the movement of pixels from one frame to another. By analyzing the changes in pixel intensities between frames, optical flow algorithms can infer the motion of objects in the scene.

Understanding the movement of objects is crucial in various computer vision tasks, including action recognition, object tracking, and, in the context of colonoscopy reconstruction, estimating the motion of the colon surface. By leveraging optical flow, researchers can extract valuable information about the dynamics and shape of the colon during the procedure.

The underlying assumption of optical flow is that objects in the scene move smoothly and continuously over time. This assumption allows the algorithm to track the apparent motion of objects by examining the pixel-level intensity changes between frames. However, optical flow

algorithms also face certain challenges and limitations due to real-world complexities.

3.2 Lucas-Kanade Method

One commonly used optical flow algorithm is the Lucas-Kanade method, which is an iterative algorithm for local motion estimation. The Lucas-Kanade method involves the following steps:

1. Selection of a small window around each pixel in the reference image.
2. Estimation of the image gradient within the window for both the reference and target images.
3. Computation of the correlation matrix between the image gradients and its inversion.
4. Multiplication of the inverted matrix by the difference between the target and reference image intensities to obtain the optical flow vector.

The Lucas-Kanade method is computationally efficient and widely used for local motion estimation tasks. However, it assumes certain conditions for accurate optical flow estimation, including brightness constancy and small motion.

Brightness constancy assumes that the brightness of an object in an image sequence does not change significantly between frames. This assumption allows the algorithm to accurately estimate the apparent motion based on intensity changes. In the case colonoscopies however, the light source moves with the camera rather than one being stationary. This means that the brightness is not exactly consistent and worsens the quality of optical flow.

Small motion assumes that objects undergo relatively small displacements between frames, ensuring the validity of optical flow vectors within a local region. For the sake of this experiment, we mostly looked at the first and last frame. Depending on the sequence at hand, there was sometimes a lot of motion. This could be mediated by simply selecting frames which are closer to the light changes so, however for our purposes, the first and last frame sufficed.

Overall, the Lucas-Kanade method serves as a foundation for optical flow estimation and provides insights into the motion dynamics of objects in image sequences.

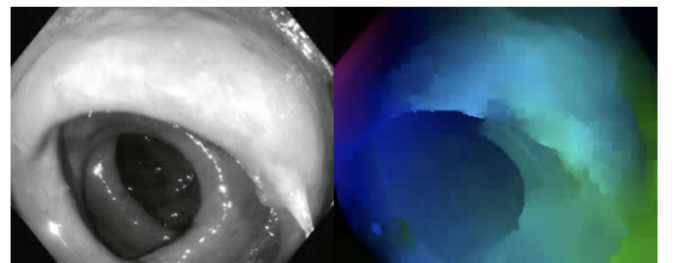


Figure 4: Optical Flow from Sequential Images On the left, we have a video of normalized values of sequential frames and on the right, we have a video of the optical flow of each pair of sequential frames.

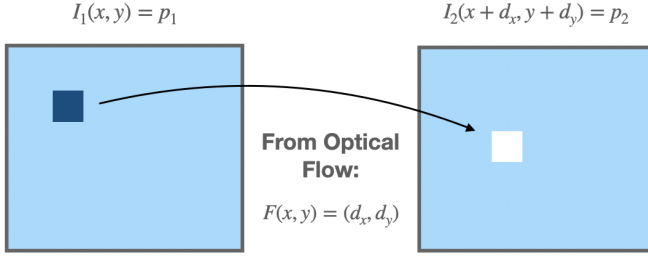


Figure 5: Optical Flow Correspondence This is a visual representation of the correspondence of two images. The left side is the original or first frame while the right side is the result or second frame. The optical flow returns a matrix with the X and Y displacement. The forward and backward methods of reconstruction are described **section 3.3**.

While its assumptions may not perfectly align with the conditions encountered in colonoscopy reconstruction, it still contributes to the understanding of colon surface motion and can be further refined to improve the accuracy and reliability of colonoscopy reconstruction algorithms.

3.3 Dense Optical Flow

Dense optical flow is a technique used to estimate the motion of every pixel in an image sequence. Unlike sparse optical flow, which estimates motion only at specific points or features, dense optical flow calculates the motion vectors for all pixels in the image.

This approach provides a detailed and comprehensive representation of the motion dynamics within the sequence. And our experiment, to compute dense optical flow we had to pre-process the data, input it into the Lucas can add method, and finally reconstruct the last frame from the resulting off to go flow two qualitatively assess the quality of the optical flow.

3.3.1 Data Pre-Processing

Before calculating optical flow, the image sequence is converted to the HSV color space. This conversion helps to separate the color and intensity components, allowing for more robust motion estimation. To enhance the reliability of the optical flow estimation, the saturation and value channels of the HSV image are normalized. This normalization step adjusts the pixel values to a standardized range, ensuring consistent intensity characteristics across frames.

By combining these steps, we are able to make the brightness of frames consistent, especially during the light change from white to blue light. Please see the video of normalized values.

3.3.2 Lucas-Kanade for Dense Optical Flow

The Lucas-Kanade method, described earlier, is applied to calculate dense optical flow. The algorithm estimates the motion vector for each pixel by analyzing the changes in pixel intensities between frames within a local neighborhood. The message takes as input to normalized images which are composed of the first and last frame. The result-

ing dense optical flow field represents the magnitude and direction of pixel motion throughout the image sequence.

3.3.3 Reconstruction from Optical Flow

In order to assess the quality of the optical flow, we are reconstructed the last frame from the optical flow. For each pixel in image 1, we extracted the ex and Y displacements from the resulting flow in order to find corresponding pixel and image 2.

We reconstructed the image in two ways: a forward and backward. Given a location in 1image 1, $I_1(x, y)$ with a pixel value of p_1 and the flow of $F(x, y)$ with displacements (d_x, d_y) we have the corresponding pixel at $I_2(x + d_x, y + d_y)$ with a value of p_2

In **forward** reconstruction we have

$$I_2(x + d_x, y + d_y) = p_1$$

In **backward** reconstruction we have

$$I_1(x, y) = p_2$$

And backward reconstruction, there are no holes in the resulting image however, and forward reconstruction, we can typically in for the missing Pixels for similar results. Please see **Figure 5** for a visual representation.

3.4 Results

The optical flow technique, along with the subsequent reconstruction of frames, yielded interesting results, which are discussed below. The effectiveness of the reconstruction process depended on the magnitude of motion within the colonoscopy videos.

In cases where there was minimal motion, such as in sequence 001_002 shown in **Figure 6a**, the reconstruction results were relatively accurate. Both the forward and backward reconstructions exhibited good fidelity. However, minor imperfections, particularly in the form of small holes, were observed in the forward reconstruction, primarily attributed to rounding errors. Conversely, the backward reconstruction appeared significantly smoother and exhibited fewer noticeable artifacts. The similarity between the first and last frames of the sequence contributed to the overall accuracy of the reconstructions.

However, when sequences with some motion were considered, as seen in **Figure 6b**, the quality of the reconstruction deteriorated rapidly. It is evident that the camera has pulled out slightly from the first frame to the last frame. In this case, numerous holes became apparent in the reconstructed frames. Although these gaps could be filled to improve the results, interpolation techniques introduce additional computational costs and may result in blocky artifacts. Despite these limitations, the backward reconstruction remained relatively acceptable, although some chunkiness was evident, along with certain sections that were clearly incorrect (e.g., the top of the backward reconstruction).

In instances where the sequences exhibited high motion, as exemplified by sequence 003_003 see **Figure 6c**, the first and last frames differed significantly. Consequently, both the forward and backward reconstructions

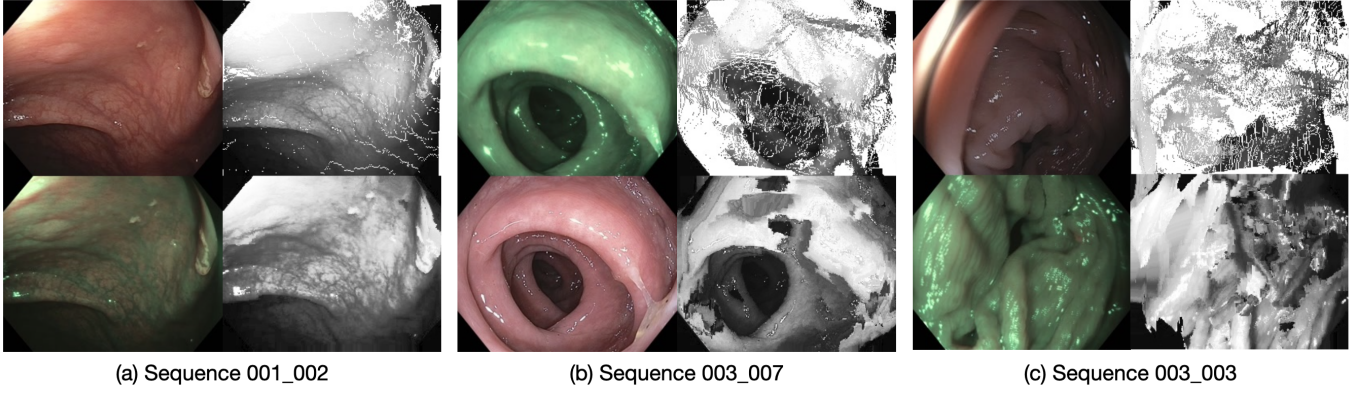


Figure 6: Optical Flow Reconstructions Each of the 3 sub figures above are composed of 4 images. The top left image is the first frame of a sequence and bottom left image is the last frame of a sequence. The top right image is the forward reconstruction and bottom right image is the backward reconstruction.

appeared scattered and lacked coherence. The reconstruction in these cases appeared relatively insignificant, highlighting the challenges associated with extreme motion scenarios. One possible approach to mitigate this issue would be to select frames closer to the actual light change, thus capturing the dynamic transitions more accurately. However, considering the goals and purposes of our research, the obtained results were deemed sufficient.

Overall, the results of the optical flow and reconstruction technique provided insights into the challenges and limitations faced in colonoscopy video analysis. While the reconstruction was generally successful in cases with minimal motion, it struggled to capture accurate representations in scenarios with higher motion. The findings underscore the importance of considering motion characteristics when selecting appropriate frames for reconstruction and suggest avenues for further improvement in handling dynamic colonoscopy sequences.

3.5 Conclusions

The dense optical flow can be utilized in various ways, including motion analysis, object tracking, and, in the case of colonoscopy reconstruction, the reconstruction of original frames. The dense optical flow information allows for the reconstruction of subsequent frames based on the displacement information provided by the flow field.

In the context of colonoscopy reconstruction, dense optical flow serves as a valuable tool for capturing the fine-grained motion of the colon surface. By analyzing the pixel-level motion vectors, researchers can gain insights into the deformations and movements of the colon during the procedure.

In conclusion, dense optical flow offers a comprehensive and detailed estimation of motion at the pixel level in image sequences. In the context of colonoscopy reconstruction, dense optical flow serves as a valuable tool for capturing the motion dynamics of the colon surface. While certain assumptions and limitations need to be considered, dense optical flow provides crucial information for reconstructing subsequent frames and understanding the deformations and movements within the colon during the procedure.

4 Other

Although it is not very exciting, the reality is that we spent weeks trying to set up environments. The RNNSLAM environment alone took many weeks and mediate the contradicting dependencies especially CUDA. This generally slowed down the process and prevented us from making having enough time to establish the final neural network.

Additionally, I have been working on implementing uflow by translating code from tensor flow to pytorch. Naturally, this is a very slow and time-consuming process given that functions do not translate one-to-one. I hope to continue to work on this project and have this implemented soon.